Silver Ground Truth Evaluation Scenarios

This appendix presents the complete input-expected output documentation for all 36 evaluation scenarios used to assess the group-based recommender system. Each scenario shows: (1) per-student input (courses, grades, interests, applications, RDIA); (2) complete ranked top-10 recommended projects; (3) Top-3 recommended interest domain, Top-3 recommended application domain, and recommended RDIA ; and (4) evaluates the alignment between the group profile and the generated recommendations. The recommender applies a balanced 50/50 scoring strategy between student competency and expressed preferences.

Scenario 1 — AI-Driven Medical Imaging and Diagnostic Support

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S001	CS1465: A CS1464: B+ CS1462: A+	Artificial Intelligence / Machine Learning Computer Vision	Healthcare / Medical Education	Health & Wellness
S002	CS1465: B CS1466: A CS1463: B+	Artificial Intelligence / Machine Learning Data Science	Healthcare / Medical Finance & Economics	Health & Wellness
S003	CS1465: C+ CS1463: B	Artificial Intelligence / Machine Learning Natural Language Processing	Healthcare / Medical	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
3	F09-42-20	Finding Deleted Regions in Human DNA
4	F29-45-20	Hudhud: AI-based Remote Rehabilitation
5	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
6	F05-45-20	Walyattawafou: Crowd Counting in Hajj and Umrah
7	F17-45-20	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
8	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
9	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters
10	F04-45-10	Baseera : Accessible Mobile App for Blind Children

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Computer Vision • Data Science	Healthcare / Medical • Education • Finance & Economics	**Health & Wellness**

The recommendations demonstrate strong alignment with the group's profile. The competency weight correctly elevates projects requiring deep neural network expertise, which matches the group's coursework in CS1465 and CS1464. The interest weight reinforces healthcare-oriented outcomes, producing a coherent and well-matched recommendation list. The RDIA category of Health & Wellness is consistently reflected in the top results.

Scenario 2 — Web-Based Intelligent Applications for Business and Education

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S004	CS1445: A CS1448: B+	Web Development Cloud Computing	Business & Productivity Education	Economies of the Future
S005	CS1446: A+ CS1447: B	Web Development Mobile Development	Business & Productivity Entertainment & Media	Economies of the Future
S006	CS1471: B+ CS1433: A	Web Development Database Systems	Business & Productivity	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F04-45-20	**CCIS Student Assistant Chatbot**
2	M05-45-20	Musanif (): A Categorization and Product Information API for E-Commerce Product Images)
3	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members
4	F03-42-10	Examining the effectiveness of pretrained language models for Arabic NLP tasks
5	F16-45-10	Automatic audio files summarization System, EJAZ Website
6	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
7	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
8	M10-45-10	School Bus Tracking and Management system
9	F06-42-20	Developing A Web Application for Automatic Code Generation
10	F04-42-10	Evaluating Website Quality Using Machine Learning

Interest Domain	Application Domain	RDIA Category
Web Development • Cloud Computing • Mobile Development	Business & Productivity • Education • Entertainment & Media	**Economies of the Future**

The recommendations are well-aligned with the group's technical and preference profile. The competency-based scoring favours projects requiring software engineering and applica- tion development skills, while the interest and application weights surface productivity and education-oriented systems. The RDIA category of Economies of the Future is consistently represented, underscoring the group's orientation toward digital transformation.

Scenario 3 — Machine Learning-Based Cyber Threat Detection

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S007	CS1439: A CS1474: B+ CS1475: A+	Cybersecurity Network Systems	Security & Privacy Government & Public Sector	Energy & Industrial Leadership
S008	CS1439: B CS1476: A	Cybersecurity Database Systems	Security & Privacy Finance & Economics	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F24-45-20	Botnet Attack Detection Using Machine Learning
3	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F02-43-10	Facial Expression Recognition for Investigations
6	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
7	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
8	F07-45-10	Secure Healthcare Information Exchange Platform KEEP SAFE
9	F05-42-10	Decentralized Social Application: Moments
10	F09-45-10	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Network Systems • Database Systems	Security & Privacy • Government & Public Sector • Finance & Economics	**Energy & Industrial Leadership**

The alignment is strong and precise. The competency weight correctly surfaces security- engineering projects aligned with the students' coursework in CS1474 and CS1476. The interest weight further reinforces cyber-focused outcomes. The RDIA category of Energy and Industrial Leadership reflects a national priority in infrastructure protection, consistent with the security application domain.

Scenario 4 — Embedded and IoT Solutions for Sustainable Smart Environments

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S009	CS1434: A CS1425: B+	Internet of Things (IoT) Embedded Systems	Sustainability & Environment Transportation & Mobility	Sustainable Environment & Supply of Essential Needs
S010	CS1424: A+ CS1463: B	Internet of Things (IoT) Robotics	Sustainability & Environment Transportation & Mobility	Energy & Industrial Leadership
S011	CS1462: B+ CS1466: A	Internet of Things (IoT) Artificial Intelligence / Machine Learning	Sustainability & Environment	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`M04-45-10`	**Smart Car Parking Manager (SCPM)**
2	`F03-45-10`	Energy-aware Multi-Object Detection and Compression in WMSN
3	`F01-45-20`	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
4	`F11-42-20`	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
5	`F10-42-20`	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs
6	`M10-45-10`	School Bus Tracking and Management system
7	`F29-45-20`	Hudhud: AI-based Remote Rehabilitation
8	`F27-45-20`	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
9	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
10	`F22-45-20`	Smart Attendance Management System Using iBeacon Technology ()

Interest Domain	Application Domain	RDIA Category
Internet of Things (IoT) • Embedded Systems • Robotics	Sustainability & Environment • Transportation & Mobility • Healthcare / Medical	**Sustainable Environment & Supply of Essential Needs**

The recommendations demonstrate excellent alignment. The competency scoring elevates projects requiring embedded and IoT engineering proficiency, while preference scoring focuses the output on sustainability and transport domains. The RDIA category of Sustain- able Environment and Supply of Essential Needs is directly reflected in the recommended project themes.

Scenario 5 — Computer Vision Applications for Medical and Assistive Contexts

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S012	CS1465: A CS1464: B CS1462: A	Computer Vision Artificial Intelligence / Machine Learning	Healthcare / Medical Education	Health & Wellness
S013	CS1465: B+ CS1466: A+	Computer Vision Data Science	Healthcare / Medical Business & Productivity	Health & Wellness
S014	CS1464: C+ CS1463: B	Computer Vision Robotics	Transportation & Mobility	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F01-42-20`	**COVID-19 Detection Using Deep Neural Networks**
2	`F01-45-20`	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
3	`F02-45-20`	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
4	`F29-45-20`	Hudhud: AI-based Remote Rehabilitation
5	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
6	`F05-45-20`	Walyattawafou: Crowd Counting in Hajj and Umrah
7	`F25-45-20`	Automatic Recognition of Children's Handwritten Arabic Characters
8	`M02-45-10`	Automated Attendance System Using Facial Recognition (FRA)
9	`F27-45-20`	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
10	`F17-45-20`	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language

Interest Domain	Application Domain	RDIA Category
Computer Vision • Artificial Intelligence / Machine Learning • Data Science	Healthcare / Medical • Education • Transportation & Mobility	**Health & Wellness**

The alignment is strong with respect to both competency and interest dimensions. However, the application match is only partial, as Transportation-oriented projects are somewhat underrepresented in the top results relative to the group's stated interest. Healthcare dominates, which is consistent but may not fully reflect the breadth of the group's application preferences.

Scenario 6 — Distributed Data Management and Context-Aware Recommendation

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S015	CS1433: A CS1426: B+	Distributed Systems Cloud Computing	Business & Productivity Finance & Economics	Economies of the Future
S016	CS1471: B CS1445: A+	Distributed Systems Database Systems	Business & Productivity Education	Economies of the Future
S017	CS1448: C+ CS1446: B+	Distributed Systems Mobile Development	Entertainment & Media	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F02-42-20`	**Online Context-Aware Recommender System**
2	`F14-42-20`	Book Recommendation System based on Collaboration Filtering (RICO)
3	`M05-45-20`	Musanif (): A Categorization and Product Information API for E-Commerce Product Images)
4	`F07-42-10`	Text Similarity Search with Application to kNN Classifier
5	`F03-42-10`	Examining the effectiveness of pretrained language models for Arabic NLP tasks
6	`F04-45-20`	CCIS Student Assistant Chatbot
7	`F16-45-10`	Automatic audio files summarization System, EJAZ Website
8	`F06-45-20`	Predicting Programming Skills Among Novice Programmers
9	`F28-45-20`	AI-trip planner and advisor
10	`M09-45-10`	Qaimni: An Interactive, Web-Based System for Rating Faculty Members

Interest Domain	Application Domain	RDIA Category
Distributed Systems • Cloud Computing • Database Systems	Business & Productivity • Finance & Economics • Education	**Economies of the Future**

Alignment is moderate-to-strong. The competency scoring correctly identifies data en- gineering and distributed systems competencies, while the preference scoring surfaces business and productivity applications. The Economies of the Future RDIA is well- represented, though Finance-specific projects are underrepresented relative to stated application priorities.

Scenario 7 — NLP-Based Systems for Education and Community Services

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S018	CS1462: A+ CS1465: A CS1466: B	Natural Language Processing Artificial Intelligence / Machine Learning	Education Social Good & Community	Health & Wellness
S019	CS1462: B+ CS1463: B	Natural Language Processing Data Science	Education Finance & Economics	Economies of the Future
S020	CS1462: C+ CS1464: A	Natural Language Processing Computer Vision	Government & Public Sector	Sustainable Environment & Supply of Essential Needs

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F03-42-10`	**Examining the effectiveness of pretrained language models for Arabic NLP tasks**
2	`F16-45-10`	Automatic audio files summarization System, EJAZ Website
3	`F06-45-10`	Get It: A Deep Learning Approach for Identifying and Discriminating Spoken Arabic Among English
4	`F17-45-20`	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
5	`F04-45-20`	CCIS Student Assistant Chatbot
6	`F09-45-10`	Detecting Cyberbullying in Twitter for Arabic Language
7	`F28-45-20`	AI-trip planner and advisor
8	`F25-45-20`	Automatic Recognition of Children's Handwritten Arabic Characters
9	`F01-43-10`	Treatment of stuttering problems in children Application - TSA
10	`F07-42-10`	Text Similarity Search with Application to kNN Classifier

Interest Domain	Application Domain	RDIA Category
Natural Language Processing • Artificial Intelligence / Machine Learning • Data Science	Education • Social Good & Community • Finance & Economics	**Health & Wellness**

The alignment is strong and highly specific. The recommendation engine correctly identifies NLP as the dominant competency and interest, producing a coherent list of language technology projects. The RDIA of Health & Wellness appears less directly reflected in the NLP context, though some results (e.g., healthcare-related text mining) bridge this gap.

Scenario 8 — Autonomous Tracking and Robotics for Vulnerable Populations

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S021	CS1424: A CS1425: B+	Robotics Embedded Systems	Transportation & Mobility Healthcare / Medical	Energy & Industrial Leadership
S022	CS1434: B CS1439: A+	Robotics Internet of Things (IoT)	Transportation & Mobility Sustainability & Environment	Sustainable Environment & Supply of Essential Needs
S023	CS1447: B+ CS1463: A	Robotics Game Development	Entertainment & Media	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F10-42-20`	**Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs**
2	`F29-45-20`	Hudhud: AI-based Remote Rehabilitation
3	`M04-45-10`	Smart Car Parking Manager (SCPM)
4	`F01-45-20`	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	`F03-45-10`	Energy-aware Multi-Object Detection and Compression in WMSN
6	`M10-45-10`	School Bus Tracking and Management system
7	`F11-42-20`	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
8	`F27-45-20`	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
9	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities

Interest Domain	Application Domain	RDIA Category
Robotics • Embedded Systems • Internet of Things (IoT)	Transportation & Mobility • Healthcare / Medical • Sustainability & Environment	**Energy & Industrial Leadership**

The alignment is highly precise. Both the competency and interest scoring components converge on robotics and IoT engineering projects. The healthcare and transportation application domains are well-reflected. The RDIA of Energy and Industrial Leadership is partially aligned, as industrial robotics is a natural extension of the group's technical profile.

Scenario 9 — Network Vulnerability Analysis and Security Intelligence

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S024	CS1474: A CS1475: B+ CS1439: A	Cybersecurity Network Systems	Security & Privacy	Energy & Industrial Leadership
S025	CS1474: B CS1476: A+	Cybersecurity Database Systems	Finance & Economics	Energy & Industrial Leadership
S026	CS1475: C+ CS1471: B	Cybersecurity Software Engineering	Government & Public Sector	Energy & Industrial Leadership

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F08-42-20`	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	`F24-45-20`	Botnet Attack Detection Using Machine Learning
3	`M02-45-20`	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	`F02-45-20`	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	`F02-43-10`	Facial Expression Recognition for Investigations
6	`F08-45-20`	XIPHER: Visualization Tool for Security Algorithms
7	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
8	`F07-45-10`	Secure Healthcare Information Exchange Platform KEEP SAFE
9	`F05-42-10`	Decentralized Social Application: Moments
10	`F09-45-10`	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Network Systems • Software Engineering	Security & Privacy • Finance & Economics • Government & Public Sector	**Energy & Industrial Leadership**

The alignment is strong and consistent. Both scoring dimensions point toward network and software security projects. However, the Finance application domain is relatively underrepresented in the recommendations compared to the group's stated priorities. The RDIA of Energy and Industrial Leadership is indirectly addressed through the infrastructure protection theme.

Scenario 10 — AI-Enabled Bioinformatics and Clinical Decision Support

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S027	CS1465: A+ CS1463: B	Artificial Intelligence / Machine Learning Bioinformatics	Healthcare / Medical	Health & Wellness
S028	CS1462: A CS1466: B+	Artificial Intelligence / Machine Learning Natural Language Processing	Education	Health & Wellness
S029	CS1464: B+ CS1424: A	Artificial Intelligence / Machine Learning Computer Vision	Transportation & Mobility	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F09-42-20	Finding Deleted Regions in Human DNA
3	F29-45-20	Hudhud: AI-based Remote Rehabilitation
4	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
6	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters
7	F04-45-10	Baseera : Accessible Mobile App for Blind Children
8	F03-42-10	Examining the effectiveness of pretrained language models for Arabic NLP tasks
9	F17-45-20	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
10	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Bioinformatics • Computer Vision	Healthcare / Medical • Education • Transportation & Mobility	**Health & Wellness**

Alignment is strong. The bioinformatics interest, which is relatively rare in the dataset, is well-captured by the recommendation engine, elevating projects such as genetic region analysis. The competency in deep learning further reinforces the AI-heavy recommendation profile. The Health & Wellness RDIA is directly and consistently reflected.

Scenario 11 — IoT-Based Monitoring Systems for Sustainable Infrastructure

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S030	CS1434: A CS1425: B+ CS1424: A+	Internet of Things (IoT) Embedded Systems	Sustainability & Environment	Sustainable Environment & Supply of Essential Needs
S031	CS1434: B CS1433: A	Internet of Things (IoT) Distributed Systems	Business & Productivity	Economies of the Future
S032	CS1434: C+ CS1471: B+	Internet of Things (IoT) Database Systems	Government & Public Sector	Energy & Industrial Leadership

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F03-45-10	**Energy-aware Multi-Object Detection and Compression in WMSN**
2	M04-45-10	Smart Car Parking Manager (SCPM)
3	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
4	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
6	M10-45-10	School Bus Tracking and Management system
7	F29-45-20	Hudhud: AI-based Remote Rehabilitation
8	F27-45-20	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
9	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
10	F10-42-20	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs

Interest Domain	Application Domain	RDIA Category
Internet of Things (IoT) • Embedded Systems • Distributed Systems	Sustainability & Environment • Business & Productivity • Government & Public Sector	**Sustainable Environment & Supply of Essential Needs**

The alignment is strong and technically precise. Both competency and interest scoring contribute to the emergence of IoT and distributed sensing projects. The sustainability and government application domains are well-represented. The RDIA of Sustainable Environment and Supply of Essential Needs is directly reflected in the top recommendations.

Scenario 12 — Platform Development for Educational and Digital Media

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S033	CS1445: A CS1448: B	Web Development Software Engineering	Entertainment & Media	Economies of the Future
S034	CS1446: B+ CS1447: A	Web Development Game Development	Education	Health & Wellness
S035	CS1471: A+ CS1433: B+	Web Development Distributed Systems	Finance & Economics	Energy & Industrial Leadership

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F04-45-20	**CCIS Student Assistant Chatbot**
2	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members
3	F06-42-20	Developing A Web Application for Automatic Code Generation
4	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
5	F04-45-10	Baseera : Accessible Mobile App for Blind Children
6	F16-45-10	Automatic audio files summarization System, EJAZ Website
7	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
8	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
9	M02-45-10	Automated Attendance System Using Facial Recognition (FRA)
10	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters

Interest Domain	Application Domain	RDIA Category
Web Development • Software Engineering • Game Development	Education • Entertainment & Media • Finance & Economics	**Economies of the Future**

The alignment is moderate-to-strong. Web development and software engineering compe- tencies are correctly identified, and the education and entertainment application domains are well-represented. However, game development projects are somewhat underrepresented in the top-10 results relative to the group's stated interest, possibly due to the limited number of game-specific projects in the project pool.

Scenario 13 — Deep Learning-Based Diagnostic Systems in Medical Imaging

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S036	CS1465: A CS1464: B+ CS1462: B	Computer Vision Artificial Intelligence / Machine Learning	Healthcare / Medical Education	Health & Wellness
S037	CS1465: B CS1466: A+	Computer Vision Data Science	Healthcare / Medical Business & Productivity	Health & Wellness
S038	CS1465: C+ CS1463: A	Data Science Bioinformatics	Healthcare / Medical	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F09-42-20	Finding Deleted Regions in Human DNA
3	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
4	F29-45-20	Hudhud: AI-based Remote Rehabilitation
5	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
6	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
7	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters
8	F05-45-20	Walyattawafou: Crowd Counting in Hajj and Umrah
9	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
10	F04-45-10	Baseera : Accessible Mobile App for Blind Children

Interest Domain	Application Domain	RDIA Category
Computer Vision • Artificial Intelligence / Machine Learning • Data Science	Healthcare / Medical • Education • Business & Productivity	**Health & Wellness**

The alignment is strong. The dual interest overlap in Computer Vision and Data Science, combined with the Healthcare application priority, produces a coherent and well-matched recommendation set. The Health & Wellness RDIA is consistently reflected. The Business and Productivity application domain is somewhat underrepresented.

Scenario 14 — Cross-Platform Application Development for Business Services

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S039	CS1445: A CS1448: B+	Web Development Cloud Computing	Business & Productivity Finance & Economics	Economies of the Future
S040	CS1446: A CS1447: B	Web Development Mobile Development	Business & Productivity	Economies of the Future
S041	CS1433: B+ CS1471: A+	Mobile Development Distributed Systems	Business & Productivity Education	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	M09-45-10	**Qaimni: An Interactive, Web-Based System for Rating Faculty Members**
2	F04-45-20	CCIS Student Assistant Chatbot
3	M05-45-20	Musanif (): A Categorization and Product Information API for E-Commerce Product Images)
4	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
5	F06-42-20	Developing A Web Application for Automatic Code Generation
6	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
7	F16-45-10	Automatic audio files summarization System, EJAZ Website
8	M10-45-10	School Bus Tracking and Management system
9	F02-42-20	Online Context-Aware Recommender System
10	F28-45-20	AI-trip planner and advisor

Interest Domain	Application Domain	RDIA Category
Web Development • Mobile Development • Cloud Computing	Business & Productivity • Finance & Economics • Education	**Economies of the Future**

The alignment is strong across both scoring dimensions. The competency weight surfaces platform development projects consistent with the group's coursework, while the prefer- ence weight correctly orients results toward business and educational applications. The Economies of the Future RDIA is consistently and appropriately reflected.

Scenario 15 — Integrated Cybersecurity and Network Forensics Systems

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S042	CS1474: A CS1439: B+ CS1475: A	Cybersecurity Network Systems	Security & Privacy Government & Public Sector	Energy & Industrial Leadership
S043	CS1474: B CS1476: A+	Cybersecurity Database Systems	Security & Privacy Finance & Economics	Sustainable Environment & Supply of Essential Needs
S044	CS1471: C+ CS1433: B	Database Systems Distributed Systems	Security & Privacy	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F24-45-20	Botnet Attack Detection Using Machine Learning
3	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F02-43-10	Facial Expression Recognition for Investigations
6	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
7	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
8	F07-45-10	Secure Healthcare Information Exchange Platform KEEP SAFE
9	F05-42-10	Decentralized Social Application: Moments
10	F09-45-10	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Network Systems • Database Systems	Security & Privacy • Government & Public Sector • Finance & Economics	**Energy & Industrial Leadership**

The alignment is strong. Both competency and interest scoring converge on security and data management projects. The inclusion of database security-specific results distinguishes this group's recommendation set from other pure-network-security groups. The Energy and Industrial Leadership RDIA is partially addressed through critical infrastructure protection themes.

Scenario 16 — Robotics and Autonomous Systems for Industrial Applications

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S045	CS1424: A CS1463: B+	Robotics Artificial Intelligence / Machine Learning	Transportation & Mobility Sustainability & Environment	Energy & Industrial Leadership
S046	CS1425: B CS1434: A+	Robotics Embedded Systems	Transportation & Mobility Healthcare / Medical	Sustainable Environment & Supply of Essential Needs
S047	CS1465: A CS1462: B+	Embedded Systems Natural Language Processing	Transportation & Mobility	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F10-42-20	**Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs**
2	F29-45-20	Hudhud: AI-based Remote Rehabilitation
3	M04-45-10	Smart Car Parking Manager (SCPM)
4	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	F03-45-10	Energy-aware Multi-Object Detection and Compression in WMSN
6	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
7	M10-45-10	School Bus Tracking and Management system
8	F27-45-20	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
9	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities

Interest Domain	Application Domain	RDIA Category
Robotics • Artificial Intelligence / Machine Learning • Embedded Systems	Transportation & Mobility • Sustainability & Environment • Healthcare / Medical	**Energy & Industrial Leadership**

The alignment is strong. The competency scoring correctly identifies robotics and embedded AI expertise, while the interest and application scoring reinforces autonomous systems and sustainability projects. The Energy and Industrial Leadership RDIA aligns with the group's interest in intelligent physical infrastructure.

Scenario 17 — Multimodal AI Systems for Healthcare and Accessibility

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S048	CS1466: A CS1465: B+ CS1464: A	Artificial Intelligence / Machine Learning Computer Vision	Healthcare / Medical Education	Health & Wellness
S049	CS1466: B CS1462: A+	Artificial Intelligence / Machine Learning Natural Language Processing	Healthcare / Medical Business & Productivity	Health & Wellness
S050	CS1466: C+ CS1463: B	Natural Language Processing Data Science	Finance & Economics	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F03-42-10	Examining the effectiveness of pretrained language models for Arabic NLP tasks
3	F29-45-20	Hudhud: AI-based Remote Rehabilitation
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F16-45-10	Automatic audio files summarization System, EJAZ Website
6	F06-45-10	Get It: A Deep Learning Approach for Identifying and Discriminating Spoken Arabic Among English
7	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
8	F17-45-20	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
9	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Computer Vision • Natural Language Processing	Healthcare / Medical • Education • Business & Productivity	**Health & Wellness**

The alignment is strong. The convergence of NLP and Computer Vision interests with a Healthcare application priority produces a recommendation set that is both technically coherent and application-relevant. The Health & Wellness RDIA is well-represented throughout the top-10 list.

Scenario 18 — Web and Game Development for Entertainment and Productivity

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S051	CS1445: A CS1448: B+	Web Development DevOps	Business & Productivity Education	Economies of the Future
S052	CS1446: B CS1447: A	Web Development Game Development	Business & Productivity Entertainment & Media	Economies of the Future
S053	CS1433: A+ CS1471: B+	Game Development Distributed Systems	Entertainment & Media Social Good & Community	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F04-45-20	**CCIS Student Assistant Chatbot**
2	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
3	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members
4	F06-42-20	Developing A Web Application for Automatic Code Generation
5	F16-45-10	Automatic audio files summarization System, EJAZ Website
6	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
7	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
8	F28-45-20	AI-trip planner and advisor
9	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
10	F04-42-10	Evaluating Website Quality Using Machine Learning

Interest Domain	Application Domain	RDIA Category
Web Development • DevOps • Game Development	Business & Productivity • Education • Entertainment & Media	**Economies of the Future**

The alignment is moderate. Web development projects are well-represented, and edu- cational applications are consistently present. However, game development projects are somewhat underrepresented in the top results, a limitation likely attributable to the relative scarcity of game development entries in the underlying project pool. The Economies of the Future RDIA is consistently reflected.

Scenario 19 — Security-Aware IoT Systems for Environmental Monitoring

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S054	CS1434: A CS1425: B+ CS1424: A	Internet of Things (IoT) Embedded Systems	Sustainability & Environment Healthcare / Medical	Sustainable Environment & Supply of Essential Needs
S055	CS1434: B CS1433: A+	Internet of Things (IoT) Distributed Systems	Sustainability & Environment Business & Productivity	Energy & Industrial Leadership
S056	CS1434: C CS1439: B	Distributed Systems Cybersecurity	Security & Privacy Business & Productivity	Economies of the Future
S057	CS1434: D+ CS1471: B	Cybersecurity Database Systems	Security & Privacy	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F03-45-10	**Energy-aware Multi-Object Detection and Compression in WMSN**
2	M04-45-10	Smart Car Parking Manager (SCPM)
3	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
4	F24-45-20	Botnet Attack Detection Using Machine Learning
5	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
6	F08-42-20	Detecting Malicious PowerShell Scripts using Machine Learning
7	F07-45-10	Secure Healthcare Information Exchange Platform KEEP SAFE
8	F29-45-20	Hudhud: AI-based Remote Rehabilitation
9	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
10	F10-42-20	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs

Interest Domain	Application Domain	RDIA Category
Internet of Things (IoT) • Embedded Systems • Distributed Systems	Sustainability & Environment • Healthcare / Medical • Security & Privacy	**Sustainable Environment & Supply of Essential Needs**

The alignment is moderate-to-strong. The overlapping interests in IoT, Distributed Systems, and Cybersecurity are all reflected in the top recommendations, though the breadth of the group's profile introduces some dispersion in results. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and consistently addressed.

Scenario 20 — Software Security and Cyber Threat Mitigation

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S058	CS1474: A+ CS1475: B	Cybersecurity Network Systems	Security & Privacy Government & Public Sector	Energy & Industrial Leadership
S059	CS1439: A CS1476: B+	Cybersecurity Software Engineering	Security & Privacy Finance & Economics	Sustainable Environment & Supply of Essential Needs
S060	CS1471: B CS1433: A+	Software Engineering Distributed Systems	Finance & Economics Business & Productivity	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F24-45-20	Botnet Attack Detection Using Machine Learning
3	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F07-45-10	Secure Healthcare Information Exchange Platform KEEP SAFE
6	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
7	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
8	F02-43-10	Facial Expression Recognition for Investigations
9	F05-42-10	Decentralized Social Application: Moments
10	F09-45-10	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Software Engineering • Network Systems	Security & Privacy • Government & Public Sector • Finance & Economics	**Energy & Industrial Leadership**

The alignment is strong. The dual emphasis on Cybersecurity and Software Engineering is appropriately captured, surfacing projects that require both development and security competencies. The Energy and Industrial Leadership RDIA is partially addressed through critical software infrastructure themes.

Scenario 21 — AI and Vision-Based Systems for Healthcare and Smart Mobility

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S061	CS1465: A CS1463: B+ CS1466: A	Artificial Intelligence / Machine Learning Data Science	Healthcare / Medical	Health & Wellness
S062	CS1465: B CS1464: A+	Artificial Intelligence / Machine Learning Computer Vision	Transportation & Mobility	Health & Wellness
S063	CS1462: B+ CS1466: C+	Computer Vision Natural Language Processing	Education	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
3	F29-45-20	Hudhud: AI-based Remote Rehabilitation
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
6	F05-45-20	Walyattawafou: Crowd Counting in Hajj and Umrah
7	F17-45-20	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
8	F25-45-20	Automatic Recognition of Children's Handwritten Arabic Characters
9	F03-42-10	Examining the effectiveness of pretrained language models for Arabic NLP tasks
10	F04-45-20	CCIS Student Assistant Chatbot

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Computer Vision • Data Science	Healthcare / Medical • Transportation & Mobility • Education	**Health & Wellness**

The alignment is strong. The shared Computer Vision and AI interests, combined with the Healthcare and Transportation application priorities, produce a well-matched set of recommendations. The Health & Wellness RDIA is directly reflected throughout the top results.

Scenario 22 — Web-Based Platforms for Entertainment and Digital Commerce

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S064	CS1445: A+ CS1448: B	Web Development Cloud Computing	Business & Productivity	Economies of the Future
S065	CS1446: A CS1447: B+	Web Development Game Development	Entertainment & Media	Economies of the Future
S066	CS1433: B CS1471: A	Game Development Database Systems	Finance & Economics	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F04-45-20	**CCIS Student Assistant Chatbot**
2	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
3	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members
4	F06-42-20	Developing A Web Application for Automatic Code Generation
5	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
6	F16-45-10	Automatic audio files summarization System, EJAZ Website
7	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
8	F28-45-20	AI-trip planner and advisor
9	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
10	F04-42-10	Evaluating Website Quality Using Machine Learning

Interest Domain	Application Domain	RDIA Category
Web Development • Cloud Computing • Game Development	Business & Productivity • Entertainment & Media • Finance & Economics	**Economies of the Future**

The alignment is moderate. Web development and business productivity projects are well-captured. Entertainment and game-related projects are somewhat underrepresented, likely due to the composition of the underlying project pool. The Economies of the Future RDIA is consistently reflected.

Scenario 23 — Embedded Robotics and IoT for Sustainability and Healthcare

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S067	CS1424: A CS1425: B+ CS1434: A	Robotics Internet of Things (IoT)	Sustainability & Environment	Energy & Industrial Leadership
S068	CS1424: B CS1463: A+	Robotics Embedded Systems	Healthcare / Medical	Sustainable Environment & Supply of Essential Needs
S069	CS1425: C CS1426: B	Embedded Systems Computer Graphics	Entertainment & Media	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F10-42-20	**Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs**
2	F29-45-20	Hudhud: AI-based Remote Rehabilitation
3	M04-45-10	Smart Car Parking Manager (SCPM)
4	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	F03-45-10	Energy-aware Multi-Object Detection and Compression in WMSN
6	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
7	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
8	F27-45-20	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
9	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	F13-42-10	Dolphin Application

Interest Domain	Application Domain	RDIA Category
Robotics • Internet of Things (IoT) • Embedded Systems	Sustainability & Environment • Healthcare / Medical • Entertainment & Media	**Energy & Industrial Leadership**

The alignment is moderate. Robotics and IoT projects are well-represented, and the healthcare and sustainability application domains are consistently captured. The Enter- tainment application dimension and the computer graphics interest are underrepresented in the recommendations, reflecting the relative scarcity of graphics-specific projects in the available pool.

Scenario 24 — Network Security and Threat Detection Engineering

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S070	CS1474: A CS1475: B+	Cybersecurity Software Engineering	Security & Privacy	Energy & Industrial Leadership
S071	CS1476: B CS1439: A+	Cybersecurity Network Systems	Government & Public Sector	Health & Wellness
S072	CS1471: A CS1433: B	Network Systems Distributed Systems	Business & Productivity	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F24-45-20	Botnet Attack Detection Using Machine Learning
3	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	F02-43-10	Facial Expression Recognition for Investigations
6	F08-45-20	XIPHER: Visualization Tool for Security Algorithms
7	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
8	F07-45-10	Secure Healthcare Information Exchange Platform KEEP SAFE
9	F05-42-10	Decentralized Social Application: Moments
10	F09-45-10	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Software Engineering • Network Systems	Security & Privacy • Government & Public Sector • Business & Productivity	**Energy & Industrial Leadership**

The alignment is strong. Both competency and interest scoring consistently elevate network security and threat intelligence projects. The Business and Productivity application dimension is less prominently reflected compared to its ranking among the group's stated priorities. The Energy and Industrial Leadership RDIA is indirectly addressed through critical network infrastructure themes.

Scenario 25 — Web and Database Systems for E-Commerce and Productivity

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S073	CS1445: A CS1448: B+ CS1433: A	Web Development DevOps	Business & Productivity Education	Economies of the Future
S074	CS1445: B CS1471: A+	Database Systems Cloud Computing	Business & Productivity Finance & Economics	Economies of the Future
S075	CS1445: C+ CS1446: B	Mobile Development Human-Computer Interaction	Business & Productivity	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	M05-45-20	**Musanif (): A Categorization and Product Information API for E-Commerce Product Images)**
2	F04-45-20	CCIS Student Assistant Chatbot
3	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
4	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members
5	F06-42-20	Developing A Web Application for Automatic Code Generation
6	F02-42-20	Online Context-Aware Recommender System
7	F16-45-10	Automatic audio files summarization System, EJAZ Website
8	F28-45-20	AI-trip planner and advisor
9	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
10	F08-45-20	XIPHER: Visualization Tool for Security Algorithms

Interest Domain	Application Domain	RDIA Category
Web Development • Cloud Computing • Database Systems	Business & Productivity • Education • Finance & Economics	**Economies of the Future**

The alignment is moderate. The diversity of interests challenges the recommender's ability to converge on a single thematic outcome, resulting in a broader but less precise recommendation set. The aggregated application preference for Business and Productivity provides a coherent anchor. The Economies of the Future RDIA is consistently reflected.

Scenario 26 — AI-Enabled Medical and Bioinformatics Research Applications

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S076	CS1465: A+ CS1463: B	Artificial Intelligence / Machine Learning Bioinformatics	Healthcare / Medical Education	Health & Wellness
S077	CS1424: B CS1425: A	Robotics Embedded Systems	Healthcare / Medical Transportation & Mobility	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F09-42-20	Finding Deleted Regions in Human DNA
3	F29-45-20	Hudhud: AI-based Remote Rehabilitation
4	F10-42-20	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs
5	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
6	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
7	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
8	M04-45-10	Smart Car Parking Manager (SCPM)
9	F03-45-10	Energy-aware Multi-Object Detection and Compression in WMSN
10	F04-45-10	Baseera : Accessible Mobile App for Blind Children

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Bioinformatics • Robotics	Healthcare / Medical • Education • Transportation & Mobility	**Health & Wellness**

The alignment is partially strong and partially weak. The AI and bioinformatics member's profile is well-captured, but the robotics and embedded systems member's competencies are less reflected in the top results. The aggregated Healthcare application preference provides cohesion, but the recommendation set may not fully serve both members' capabilities equally.

Scenario 27 — IoT-Driven Environmental Sensing and Smart Data Systems

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S078	CS1434: A CS1425: B+ CS1433: A	Internet of Things (IoT) Cloud Computing	Sustainability & Environment Business & Productivity	Sustainable Environment & Supply of Essential Needs
S079	CS1434: B CS1471: A+	Database Systems Distributed Systems	Sustainability & Environment Finance & Economics	Economies of the Future
S080	CS1434: C+ CS1462: B	Natural Language Processing Bioinformatics	Sustainability & Environment	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F03-45-10`	**Energy-aware Multi-Object Detection and Compression in WMSN**
2	`M04-45-10`	Smart Car Parking Manager (SCPM)
3	`F11-42-20`	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
4	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
5	`F01-45-20`	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
6	`F02-42-20`	Online Context-Aware Recommender System
7	`F07-45-10`	Secure Healthcare Information Exchange Platform KEEP SAFE
8	`F29-45-20`	Hudhud: AI-based Remote Rehabilitation
9	`F16-45-10`	Automatic audio files summarization System, EJAZ Website
10	`F10-42-20`	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs

Interest Domain	Application Domain	RDIA Category
Internet of Things (IoT) • Cloud Computing • Distributed Systems	Sustainability & Environment • Business & Productivity • Finance & Economics	**Sustainable Environment & Supply of Essential Needs**

The alignment is moderate. The IoT and sustainability connection is well-captured, while the NLP and bioinformatics interests are less directly reflected. The distributed systems and cloud computing competencies contribute to the systems-level recommendation outputs. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and consistently reflected.

Scenario 28 — Intelligent Threat Detection with AI and Security Engineering

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S081	CS1474: A CS1475: B+	Cybersecurity Software Engineering	Security & Privacy Government & Public Sector	Energy & Industrial Leadership
S082	CS1465: A+ CS1463: B	Artificial Intelligence / Machine Learning Data Science	Security & Privacy	Economies of the Future

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F08-42-20`	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	`F24-45-20`	Botnet Attack Detection Using Machine Learning
3	`M02-45-20`	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	`F02-45-20`	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
6	`F02-43-10`	Facial Expression Recognition for Investigations
7	`F08-45-20`	XIPHER: Visualization Tool for Security Algorithms
8	`F07-45-10`	Secure Healthcare Information Exchange Platform KEEP SAFE
9	`F05-42-10`	Decentralized Social Application: Moments
10	`F09-45-10`	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Artificial Intelligence / Machine Learning • Software Engineering	Security & Privacy • Government & Public Sector • Data Science	**Energy & Industrial Leadership**

The alignment is strong and well-balanced across both members. The recommendation engine successfully identified the AI-cybersecurity intersection as the primary output domain, producing results that require contributions from both competency sets. The Energy and Industrial Leadership RDIA is appropriately reflected in the critical security infrastructure theme.

Scenario 29 — Comprehensive Network Security and Governance Systems

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S083	CS1439: A CS1476: B+ CS1474: A	Cybersecurity Network Systems	Security & Privacy Government & Public Sector	Energy & Industrial Leadership
S084	CS1439: B CS1475: A+	Software Engineering Cloud Computing	Security & Privacy Business & Productivity	Energy & Industrial Leadership
S085	CS1439: C+ CS1471: B	Database Systems Distributed Systems	Finance & Economics	Energy & Industrial Leadership

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F08-42-20`	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	`F24-45-20`	Botnet Attack Detection Using Machine Learning
3	`M02-45-20`	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	`F02-45-20`	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
5	`F02-43-10`	Facial Expression Recognition for Investigations
6	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
7	`F07-45-10`	Secure Healthcare Information Exchange Platform KEEP SAFE
8	`F08-45-20`	XIPHER: Visualization Tool for Security Algorithms
9	`F05-42-10`	Decentralized Social Application: Moments
10	`F09-45-10`	Detecting Cyberbullying in Twitter for Arabic Language

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Network Systems • Software Engineering	Security & Privacy • Government & Public Sector • Business & Productivity	**Energy & Industrial Leadership**

The alignment is moderate-to-strong. The cybersecurity and network systems competencies are well-captured in the top results. Cloud computing and distributed systems are less prominently represented, though back-end security and infrastructure management projects partially address these interests. The Energy and Industrial Leadership RDIA is indirectly addressed.

Scenario 30 — AI and Computer Vision for Clinical and Social Applications

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S086	CS1465: A+ CS1463: B	Artificial Intelligence / Machine Learning Computer Vision	Healthcare / Medical Education	Sustainable Environment & Supply of Essential Needs
S087	CS1462: A CS1466: B+	Natural Language Processing Data Science	Healthcare / Medical Social Good & Community	Sustainable Environment & Supply of Essential Needs
S088	CS1434: B+ CS1425: A	Internet of Things (IoT) Embedded Systems	Transportation & Mobility	Sustainable Environment & Supply of Essential Needs

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F01-42-20`	**COVID-19 Detection Using Deep Neural Networks**
2	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
3	`F29-45-20`	Hudhud: AI-based Remote Rehabilitation
4	`F03-45-10`	Energy-aware Multi-Object Detection and Compression in WMSN
5	`F01-45-20`	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
6	`F17-45-20`	Munis: Enhancing Deaf Communication with Realtime Subtitles in Arabic Language
7	`F09-45-10`	Detecting Cyberbullying in Twitter for Arabic Language
8	`M04-45-10`	Smart Car Parking Manager (SCPM)
9	`F10-42-20`	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs
10	`F25-45-20`	Automatic Recognition of Children's Handwritten Arabic Characters

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Computer Vision • Natural Language Processing	Healthcare / Medical • Education • Social Good & Community	**Sustainable Environment & Supply of Essential Needs**

The alignment is moderate. The AI and healthcare dimensions are well-captured, while IoT and embedded systems competencies are less prominently represented in the top results. The Social Good and Community application domain is partially reflected in some NLP and accessibility-oriented recommendations. The RDIA of Sustainable Environment and Supply of Essential Needs is moderately represented.

Scenario 31 — Human-Centered Web and Interactive Application Design

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S089	CS1445: A CS1448: B+ CS1446: A	Web Development Human-Computer Interaction	Education Social Good & Community	Economies of the Future
S090	CS1445: B CS1447: A+	Game Development Computer Graphics	Education Entertainment & Media	Health & Wellness
S091	CS1445: C CS1471: B	Database Systems Cloud Computing	Business & Productivity	Energy & Industrial Leadership
S092	CS1445: D+ CS1433: A	Distributed Systems DevOps	Government & Public Sector	Sustainable Environment & Supply of Essential Needs

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`M09-45-10`	**Qaimni: An Interactive, Web-Based System for Rating Faculty Members**
2	`F04-45-20`	CCIS Student Assistant Chatbot
3	`F14-42-20`	Book Recommendation System based on Collaboration Filtering (RICO)
4	`F04-45-10`	Baseera : Accessible Mobile App for Blind Children
5	`F06-42-20`	Developing A Web Application for Automatic Code Generation
6	`F08-45-20`	XIPHER: Visualization Tool for Security Algorithms
7	`F22-45-20`	Smart Attendance Management System Using iBeacon Technology ()
8	`F16-45-10`	Automatic audio files summarization System, EJAZ Website
9	`F28-45-20`	AI-trip planner and advisor
10	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities

Interest Domain	Application Domain	RDIA Category
Web Development • Human-Computer Interaction • Game Development	Education • Social Good & Community • Entertainment & Media	**Economies of the Future**

The alignment is moderate. The recommendation engine successfully identifies the ed- ucation and social good application cluster as the coherent output domain, despite the extreme interest diversity. Individual interest areas such as DevOps, computer graphics, and cloud computing are only partially captured. The Economies of the Future RDIA is consistently reflected.

Scenario 32 — Hybrid AI-Cybersecurity Solutions for Connected Environments

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S093	CS1474: A CS1475: B+	Cybersecurity Network Systems	Security & Privacy Finance & Economics	Energy & Industrial Leadership
S094	CS1465: A CS1463: B	Artificial Intelligence / Machine Learning Bioinformatics	Security & Privacy Healthcare / Medical	Health & Wellness
S095	CS1433: B+ CS1434: A+	Internet of Things (IoT) Distributed Systems	Sustainability & Environment	Sustainable Environment & Supply of Essential Needs

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	`F08-42-20`	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	`F24-45-20`	Botnet Attack Detection Using Machine Learning
3	`M02-45-20`	Detection of Adversarial Attacks Malwares Using Machine Learning Models
4	`F01-42-20`	COVID-19 Detection Using Deep Neural Networks
5	`F09-42-20`	Finding Deleted Regions in Human DNA
6	`F02-45-20`	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
7	`F07-45-10`	Secure Healthcare Information Exchange Platform KEEP SAFE
8	`F20-45-20`	Abnormal Event Detection for Video Surveillance (SecuritEye)
9	`F03-45-10`	Energy-aware Multi-Object Detection and Compression in WMSN
10	`F05-45-10`	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Artificial Intelligence / Machine Learning • Internet of Things (IoT)	Security & Privacy • Healthcare / Medical • Sustainability & Environment	**Energy & Industrial Leadership**

The alignment is strong for the cybersecurity and AI dimensions but moderate for the IoT, bioinformatics, and sustainability dimensions. The recommendation engine successfully identifies the AI-cybersecurity intersection as the primary output domain. The Energy and Industrial Leadership RDIA is appropriately and consistently reflected.

Scenario 33 — Vision-Based AI for Healthcare and Autonomous Navigation

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S096	CS1465: A CS1464: B+ CS1462: A	Artificial Intelligence / Machine Learning Computer Vision	Healthcare / Medical	Health & Wellness
S097	CS1465: B CS1424: A+	Robotics Embedded Systems	Transportation & Mobility	Health & Wellness
S098	CS1465: C+ CS1471: B	Database Systems Cloud Computing	Business & Productivity	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F01-42-20	**COVID-19 Detection Using Deep Neural Networks**
2	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
3	F10-42-20	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs
4	F29-45-20	Hudhud: AI-based Remote Rehabilitation
5	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
6	M04-45-10	Smart Car Parking Manager (SCPM)
7	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
8	F05-45-20	Walyattawafou: Crowd Counting in Hajj and Umrah
9	F03-45-10	Energy-aware Multi-Object Detection and Compression in WMSN
10	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)

Interest Domain	Application Domain	RDIA Category
Artificial Intelligence / Machine Learning • Computer Vision • Robotics	Healthcare / Medical • Transportation & Mobility • Business & Productivity	**Health & Wellness**

The alignment is moderate. Healthcare AI projects are well-captured for the first member, while robotics and embedded systems projects are partially addressed in the mid-tier recommendations. Database and cloud computing competencies are less prominently reflected. The Health & Wellness RDIA is consistently present.

Scenario 34 — Intelligent Cybersecurity with AI and Web Integration

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S099	CS1439: A+ CS1474: B	Cybersecurity Network Systems	Security & Privacy	Energy & Industrial Leadership
S100	CS1445: A CS1448: B+	Web Development DevOps	Business & Productivity	Energy & Industrial Leadership
S101	CS1465: B CS1463: A	Artificial Intelligence / Machine Learning Bioinformatics	Healthcare / Medical	Energy & Industrial Leadership

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F01-42-20	COVID-19 Detection Using Deep Neural Networks
3	F24-45-20	Botnet Attack Detection Using Machine Learning
4	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
5	F09-42-20	Finding Deleted Regions in Human DNA
6	M05-45-20	Musanif (): A Categorization and Product Information API for E-Commerce Product Images)
7	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
8	F04-45-20	CCIS Student Assistant Chatbot
9	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	M09-45-10	Qaimni: An Interactive, Web-Based System for Rating Faculty Members

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Artificial Intelligence / Machine Learning • Web Development	Security & Privacy • Healthcare / Medical • Business & Productivity	**Energy & Industrial Leadership**

The alignment is strong for the security and AI dimensions. The web development and DevOps interests are partially captured through security platform and dashboard recommendations. The bioinformatics interest is less directly addressed. The Energy and Industrial Leadership RDIA is consistently and appropriately reflected.

Scenario 35 — Embedded IoT and Drone Systems for Environment and Transport

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S102	CS1434: A CS1425: B+ CS1433: A	Internet of Things (IoT) Embedded Systems	Sustainability & Environment	Sustainable Environment & Supply of Essential Needs
S103	CS1434: B CS1424: A+	Robotics Drones	Transportation & Mobility	Energy & Industrial Leadership
S104	CS1434: C CS1462: B	Natural Language Processing Data Science	Education	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F03-45-10	**Energy-aware Multi-Object Detection and Compression in WMSN**
2	M04-45-10	Smart Car Parking Manager (SCPM)
3	F10-42-20	Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs
4	F01-45-20	Baasyir:A YOLO Based approach to Assist Visually Impaired People by leveraging Smart Glasses System
5	F11-42-20	Energy-Efficient Data Reduction Techniques for Sensor-based ECG Monitoring Applications
6	F05-45-10	Visualize: A Software Tool for Visualizing the Impact of Climate Change on Cities
7	F29-45-20	Hudhud: AI-based Remote Rehabilitation
8	F16-45-10	Automatic audio files summarization System, EJAZ Website
9	F27-45-20	Maseer: Advanced Pilgrimage Mapping for Effective Crowd Management during Hajj
10	F03-42-10	Examining the effectiveness of pretrained language models for Arabic NLP tasks

Interest Domain	Application Domain	RDIA Category
Internet of Things (IoT) • Embedded Systems • Robotics	Sustainability & Environment • Transportation & Mobility • Education	**Sustainable Environment & Supply of Essential Needs**

The alignment is moderate-to-strong. The IoT and drone components are well-captured in the recommendations, while the NLP and data science interests are less prominently reflected. The sustainability and transport application priorities are consistently addressed. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and appropriately reflected.

Scenario 36 — Cross-Domain AI and Cybersecurity Application Development

Group:

INPUT — Student Group Profile

Student	Courses & Grades	Interests	Applications	RDIA
S105	CS1474: A+ CS1475: B	Cybersecurity Software Engineering	Security & Privacy	Energy & Industrial Leadership
S106	CS1445: A CS1446: B+	Web Development Mobile Development	Entertainment & Media	Economies of the Future
S107	CS1465: B+ CS1463: A	Artificial Intelligence / Machine Learning Bioinformatics	Healthcare / Medical	Health & Wellness

EXPECTED OUTPUT — Recommendation

#	Project ID	Project Title
1	F08-42-20	**Detecting Malicious PowerShell Scripts using Machine Learning**
2	F01-42-20	COVID-19 Detection Using Deep Neural Networks
3	F24-45-20	Botnet Attack Detection Using Machine Learning
4	M02-45-20	Detection of Adversarial Attacks Malwares Using Machine Learning Models
5	F09-42-20	Finding Deleted Regions in Human DNA
6	F14-42-20	Book Recommendation System based on Collaboration Filtering (RICO)
7	F02-45-20	DeepFake Videos Detection Based on A Hybrid Model Using Deep Learning
8	F22-45-20	Smart Attendance Management System Using iBeacon Technology ()
9	F20-45-20	Abnormal Event Detection for Video Surveillance (SecuritEye)
10	F16-45-10	Automatic audio files summarization System, EJAZ Website

Interest Domain	Application Domain	RDIA Category
Cybersecurity • Artificial Intelligence / Machine Learning • Web Development	Security & Privacy • Healthcare / Medical • Entertainment & Media	**Energy & Industrial Leadership**

The alignment is strong for the security and AI dimensions, but moderate for the web, mobile, and bioinformatics dimensions. The entertainment application preference is underrepresented relative to the healthcare and security themes that dominate the top-10 results. The Energy and Industrial Leadership RDIA is consistently reflected.